

T.H.V.K. Nilaweera

Electrical & Electronic Engineering Undergraduate | Automation & Embedded Systems

+94 71 990 0468 | +94 72 723 1067 | Nilaweeravenuajcam@gmail.com | Gampaha, Sri Lanka
[linkedin.com/in/venuja-nilaweera](https://www.linkedin.com/in/venuja-nilaweera) | github.com/VenujaNilaweera

PROFESSIONAL SUMMARY

Final-year Electrical and Electronic Engineering undergraduate seeking an embedded systems and IoT automation internship. Proven track record carrying multiple projects end to end, from initial concept through SolidWorks CAD design, PCB design and embedded firmware, to fully working, demonstrated prototypes, each documented on video and exhibited nationally.

EDUCATION

University of Jaffna

2022 to 2026 (Expected)

BSc Eng. (Hons) in Electrical and Electronic Engineering, Faculty of Engineering

President's College, Minuwangoda

GCE Advanced Level: 2A, 1C | GCE Ordinary Level: 9A

SKILLS

- **Embedded Systems and IoT:** ESP32 (Wi-Fi enabled), Arduino, Arduino Nano, real-time firmware design, I2C/SPI/UART sensor interfacing, stepper, DC and servo motor control, sensor integration (inertial, thermal, infrared, optical, pressure)
- **Control Systems:** PID (Proportional-Integral-Derivative) control and controller tuning, closed-loop feedback, real-time sensor fusion, waypoint navigation
- **Mechanical Design (CAD):** SolidWorks (4 years), part and assembly modelling, mechanism and linkage design, design for 3D printing and retrofit integration
- **Electronics Design:** Altium Designer, PCB (Printed Circuit Board) layout, schematic capture
- **Computer Vision and ML:** OpenCV, YOLOv8 object detection, Edge AI
- **Programming and Tools:** Python, C/C++, MATLAB, Git, GitHub, Visual Studio Code

PROJECTS

Each taken from concept through SolidWorks CAD (and PCB design where applicable) to a fully working, demonstrated prototype – video demonstrations linked below.

AUTOBOT-X: Autonomous Mobile Manipulation Robot

2026

Autonomous Navigation, PID Motion Control and Robotic Pick-and-Place

- Built a fully working autonomous mobile robot on ESP32 firmware with custom PCB and a 100 Hz sensor loop, running tuned closed-loop PID motion control for waypoint navigation and precision turning.
- Integrated a multi-element infrared array and filtered colour sensor for real-time obstacle detection, target identification, and dynamic collision avoidance.
- Designed a custom servo-actuated gripper in SolidWorks, modelling the linkage mechanism for precision grasping and coordinated transport.

Tech Stack: ESP32, PID Control and Tuning, IR Sensor Array, Colour Sensor, Servo Actuation, Altium PCB, SolidWorks

ThermalFlow: Thermal Occupancy Intelligence Platform

Patent Filing in Progress | 2023 to Present

Developer | Occupancy-Aware Control and Edge Processing

[Watch on YouTube](#)

- Designed and built a privacy-preserving occupancy detection platform using a 16x12 MLX90641 infrared thermal array and an ESP32, implementing a tuned PID control loop to drive occupancy-aware actuation of connected cooling equipment.
- Developed on-device algorithms for ambient temperature compensation, thermal noise filtering, and non-human heat source rejection, enabling real-time multi-occupant tracking entirely on the edge.
- Modelled sensor housings in SolidWorks, ran extended residential testing, and presented at Techno Exhibition 2025 (BMICH) before advancing to IEEE Innovation Nation Sri Lanka 2026.

Tech Stack: ESP32, MLX90641 Thermal Array, Embedded C/C++, PID Control and Tuning, SolidWorks, Edge AI

Automated Parasitic Egg Detection System

Patent Filing in Progress | 2025

Motion-Controlled Scanning Automation and Machine Vision Inspection

[Watch on YouTube](#)

- Automated microscope slide scanning using NEMA 17 stepper motors and A4988 drivers on an Arduino Nano, implementing zig-zag traversal, grid dimension calculation, and home-position return.
- Modelled and 3D-printed the full mechanical assembly in SolidWorks (X/Y slide stage, motor mounting frame, and eyepiece-to-camera adapter) as a retrofit onto an existing laboratory microscope.
- Trained and deployed a YOLOv8 model with an OpenCV pipeline to detect and count roundworm, hookworm and whipworm eggs in real time, matching Medical Laboratory Technologist counts to within 1 to 4 eggs per slide across six validated stool samples.

Tech Stack: Arduino Nano, NEMA 17 Steppers, A4988 Drivers, SolidWorks, 3D Printing, Python, OpenCV, YOLOv8

Project ASW: Active Suspension Wheelchair

2024 to 2025

Mechatronic Design, Closed-Loop Leveling Control and Fabrication

[Watch on YouTube](#)

- Engineered a retrofit active-suspension system, designing the mechanism and actuator mounting geometry in SolidWorks for approximately 15 cm of independent left and right terrain compensation.
- Implemented and tuned closed-loop PID leveling controllers using three MPU6050 inertial measurement units and position feedback to drive two DC geared motor and lead-screw actuators independently, adjusting gains for smooth, drift-free leveling response.
- Fabricated the prototype end to end through cutting, welding and assembly, and demonstrated it at Techno 2025.

Tech Stack: SolidWorks, PID Control and Tuning, MPU6050 IMU, DC Geared Motors, Lead Screws, Metal Fabrication

FINAL-YEAR RESEARCH

Wearable Biomedical System for Dengue Critical Phase Detection

2026 to Present

Undergraduate Research Project, Team Member | Department of Electrical and Electronic Engineering

- Developing a wearable system to continuously monitor confirmed dengue patients and flag the critical-phase transition earlier than periodic clinical blood tests, fusing PPG heart rate, oscillometric blood pressure, skin temperature, and automated Capillary Refill Time (CRT).
- Engineered a novel automated CRT actuation module – a pneumatic pump-solenoid pressure system (four Arduino-driven IRFZ44N MOSFETs) driving a syringe-based fingertip actuator for repeatable, operator-independent nail-bed compression and release – paired with an automated image-processing pipeline (LAB colour-space conversion, watershed segmentation, ellipse-based ROI refinement) that tracks the a* channel colour-recovery curve to extract CRT.
- Developed a wrist-worn wearable module with a custom-designed PCB integrating PPG and dual axillary temperature sensors, and validated the CRT response against literature patterns and a clinical perception study with medical personnel.

Tech Stack: Arduino, Python, OpenCV, Pneumatic Actuation, MOSFET Drivers, PCB Design, LAB Colour Space, Wearable Sensors

CERTIFICATIONS AND COURSES

- **PCB Basic Design Course** - Altium Education, August 2026
- **Embedded Machine Learning (Edge AI Fundamentals)** - Edge Impulse via Coursera, August 2026
- **Python for Beginners and Web Design for Beginners** - University of Moratuwa (CODL), 2022

ACTIVITIES AND ACHIEVEMENTS

- **Patent Filings in Progress** for ThermalFlow and the Automated Parasitic Egg Detection System.
- **Exhibitor**, Techno Exhibition 2025, BMICH Colombo. Showcased two independently built prototypes.
- **Student Member**, Institution of Engineering and Technology (IET), University of Jaffna, 2024.

REFERENCES

Prof. Tharmarajah Thiruvaran

Head, Dept. of EEE, University of Jaffna

thiruvaran@eng.jfn.ac.lk | +94 21 206 0160

Dr. Tharmakulasingam Mukunthan

Senior Lecturer, Research Supervisor, Dept. of EEE

mukunthan@eng.jfn.ac.lk | +94 21 206 0161